assignment2

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0.1 Statistical Analysis

0.2 Assignment 2

0.3 PRATIK SUNIL ABHANG

0.4 260740150018

[1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

[2]:
```
df = pd.read_csv(r"C:\Users\prati\Downloads\diabetes.csv")

df.head()
```

[2]:

	id	chol	stab.glu	hdl	ratio	glyhb	location	age	gender	height	\
0	1000	203.0	82	56.0	3.6	4.31	Buckingham	46	female	62.0	
1	1001	165.0	97	24.0	6.9	4.44	Buckingham	29	female	64.0	
2	1002	228.0	92	37.0	6.2	4.64	Buckingham	58	female	61.0	
3	1003	78.0	93	12.0	6.5	4.63	Buckingham	67	male	67.0	
4	1005	249.0	90	28.0	8.9	7.72	Buckingham	64	male	68.0	

	weight	frame	bp.1s	bp.1d	bp.2s	bp.2d	waist	hip	time.ppn
0	121.0	medium	118.0	59.0	NaN	NaN	29.0	38.0	720.0
1	218.0	large	112.0	68.0	NaN	NaN	46.0	48.0	360.0
2	256.0	large	190.0	92.0	185.0	92.0	49.0	57.0	180.0
3	119.0	large	110.0	50.0	NaN	NaN	33.0	38.0	480.0
4	183.0	medium	138.0	80.0	NaN	NaN	44.0	41.0	300.0

[3]:
```
df.isnull().sum()
```

[3]:
```
id            0
chol          1
stab.glu      0
hdl           1
ratio         1
glyhb        13
location      0
age           0
```

```
gender        0
height        5
weight        1
frame        12
bp.1s         5
bp.1d         5
bp.2s       262
bp.2d       262
waist         2
hip           2
time.ppn      3
dtype: int64
```

Median imputation for BP-related columns

[4]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 403 entries, 0 to 402
Data columns (total 19 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   id        403 non-null    int64
 1   chol      402 non-null    float64
 2   stab.glu  403 non-null    int64
 3   hdl       402 non-null    float64
 4   ratio     402 non-null    float64
 5   glyhb     390 non-null    float64
 6   location  403 non-null    str
 7   age       403 non-null    int64
 8   gender    403 non-null    str
 9   height    398 non-null    float64
 10  weight    402 non-null    float64
 11  frame     391 non-null    str
 12  bp.1s     398 non-null    float64
 13  bp.1d     398 non-null    float64
 14  bp.2s     141 non-null    float64
 15  bp.2d     141 non-null    float64
 16  waist     401 non-null    float64
 17  hip       401 non-null    float64
 18  time.ppn  400 non-null    float64
dtypes: float64(13), int64(3), str(3)
memory usage: 67.3 KB
```

[5]:
```
df.columns
```

[5]:
```
Index(['id', 'chol', 'stab.glu', 'hdl', 'ratio', 'glyhb', 'location', 'age',
       'gender', 'height', 'weight', 'frame', 'bp.1s', 'bp.1d', 'bp.2s',
```

```
       'bp.2d', 'waist', 'hip', 'time.ppn'],
      dtype='str')
```

Median imputation for BP columns

[6]:
```
df['bp.1s'] = df['bp.1s'].fillna(df['bp.1s'].median())

df['bp.1d'] = df['bp.1d'].fillna(df['bp.1d'].median())

df['bp.2s'] = df['bp.2s'].fillna(df['bp.2s'].median())

df['bp.2d'] = df['bp.2d'].fillna(df['bp.2d'].median())
```

Mean imputation for all other numerical columns

[7]:
```
df['chol'] = df['chol'].fillna(df['chol'].mean())

df['stab.glu'] = df['stab.glu'].fillna(df['stab.glu'].mean())

df['hdl'] = df['hdl'].fillna(df['hdl'].mean())

df['ratio'] = df['ratio'].fillna(df['ratio'].mean())

df['glyhb'] = df['glyhb'].fillna(df['glyhb'].mean())

df['age'] = df['age'].fillna(df['age'].mean())

df['height'] = df['height'].fillna(df['height'].mean())

df['weight'] = df['weight'].fillna(df['weight'].mean())

df['waist'] = df['waist'].fillna(df['waist'].mean())

df['hip'] = df['hip'].fillna(df['hip'].mean())

df['time.ppn'] = df['time.ppn'].fillna(df['time.ppn'].mean())
```

[8]:
```
df.isnull().sum()
```

[8]:
```
id           0
chol         0
stab.glu     0
hdl          0
ratio        0
glyhb        0
location     0
age          0
gender       0
```

```
height       0
weight       0
frame       12
bp.1s        0
bp.1d        0
bp.2s        0
bp.2d        0
waist        0
hip          0
time.ppn     0
dtype: int64
```

Crosstab – Location with Gender

[9]:
```
pd.crosstab(df['location'], df['gender'])
```

[9]:
```
gender      female  male
location
Buckingham     114    86
Louisa         120    83
```

Crosstab – Gender with Frame

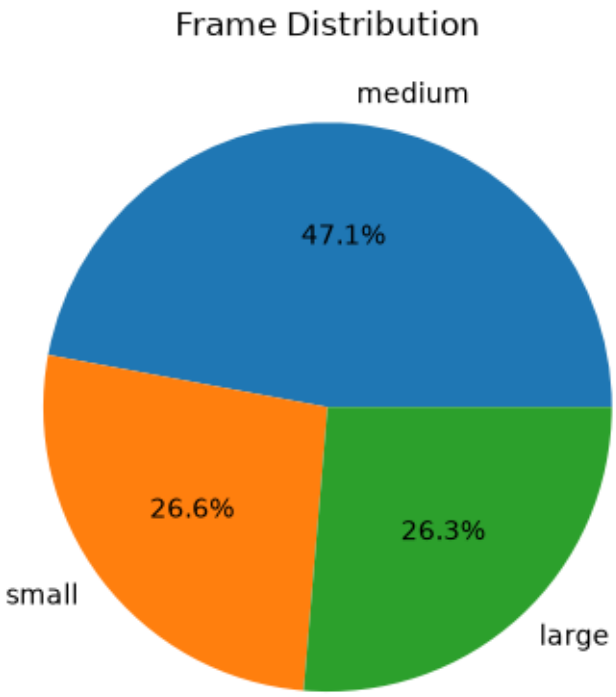
[10]:
```
pd.crosstab(df['gender'], df['frame'])
```

[10]:
```
frame   large  medium  small
gender
female     42     116     69
male       61      68     35
```

Pie chart – Frame

[11]:
```
df['frame'].value_counts().plot(kind='pie', autopct='%1.1f%%')

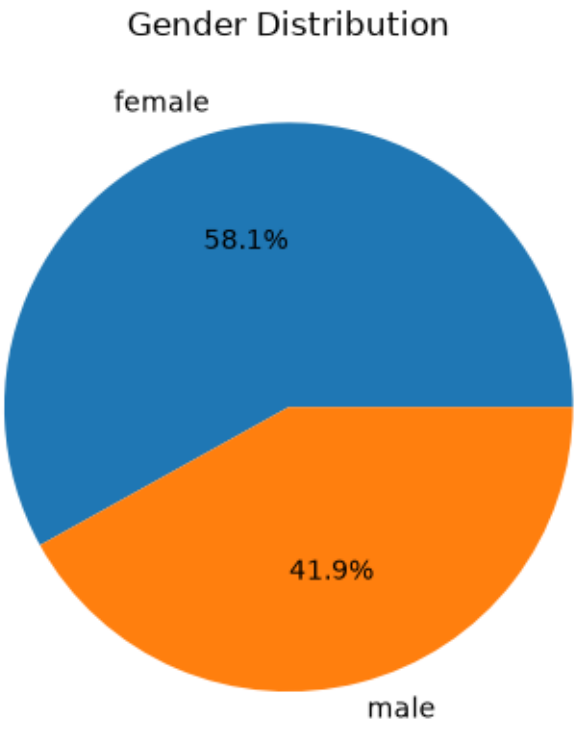
plt.title('Frame Distribution')
plt.ylabel('')
plt.show()
```



Pie chart – Gender

[12]:
```
df['gender'].value_counts().plot(kind='pie', autopct='%1.1f%%')

plt.title('Gender Distribution')
plt.ylabel('')
plt.show()
```



[13]:
```
df.to_csv(r"C:\Users\prati\Downloads\diabetes_imputed.csv", index=False)

print("Dataset saved successfully")
```

```
Dataset saved successfully
```

[ ]:
```

```